Normalization :

- Normalization is a core database design principle, not specific to MySQL, used to organize data efficiently to reduce redundancy and improve data integrity.
- This is achieved by systematically breaking down large tables into smaller, related tables and defining relationships between them using keys (primary and foreign).

Normalization is achieved through a series of **"Normal Forms" (NF)**, each with specific rules that build upon the last. The most common are:

First Normal Form (1NF):

The table must have atomic values in each column, and there should be no repeating groups of data.

Each row must be unique, typically identified by a primary key.

Second Normal Form (2NF):

The table must be in 1NF, and all non-key attributes must be fully functionally

Third Normal Form (3NF):

The table must be in 2NF
Removes transitive dependencies.

First Normal form(1NF) :
Each column must contain only a single, atomic value and there should be no repeating groups of data.

Example

userid	fullname	phone_numbers
1	John Doe	123-4567, 890-1234
2	Jane Smith	555-6789

After Normalization:

userid	firstname	lastname
1	John	Doe
2	Jane	Smith

Second Normal Form (2NF)

A table must be in 1NF, and all non-key attributes must be fully dependent on the entire primary key. This typically applies to tables with a composite primary key.

StudentID	ProjectNo	StudentName	ProjectName
S001	P1	Alice	Database Design
S001	P2	Alice	MySQL Basics
S002	P1	Bob	Database Design

StudentID	StudentName
S001	Alice
S002	Bob

projects table:

ProjectNo	ProjectName
P1	Database Design
P2	MySQL Basics

student_project_enrollment table:

StudentID	ProjectNo
S001	P1
S001	P2
S002	P1

Third Normal Form (3NF)

A table must be in 2NF, and there should be no transitive dependencies. This means non-key attributes should not depend on other non-key attributes.

EmployeeID	Name	JobCode	JobTitle
E001	Alice	J01	Chef
E002	Bob	J02	Waiter
E003	Charlie	J01	Chef

Normalized Example (3NF):

EmployeeID	Name	JobCode
E001	Alice	J01
E002	Bob	J02
E003	Charlie	J01

jobs table:

JobCode	JobTitle
J01	Chef
J02	Waiter

Normalization is an essential database design technique that ensures:

- Clean structure
- Efficient storage
- Reliable data handling